Appendix: LP Estimation Variants for Part 2

`Q_Part2_NoVIX.m` (v1) vs `v2.m` vs `v3.m`

1. Why three variants?

The three scripts implement the *same* candidate-mechanism level LP but differ in how the regression is run, how the HAC standard errors are formed, and (in v3 only) which lag blocks enter the design matrix. This appendix isolates those differences so that the IRF discrepancies across versions can be attributed to specific design choices rather than to unmodelled drift.

2. LP specification

Common settings across v1/v2/v3: $p_{\text{innov}} = 1$, $p_{\text{lp}} = 4$, $H = 12$, CI $= 90\,\%$, shock size $= 1\sigma$, controls $= [\text{Unemp}, \text{FFR}]$, no VIX in controls. Shock construction (univariate AR(1) residual on the log GPR level, in-sample z-scored) is also identical across all three.

2.1. v1 and v2

For each outcome y, each shock series $z^{(s)} \in \{z^{\text{World}}, z^{\text{Threat}}, z^{\text{Act}}\}$, and each horizon $h \in \{0, \dots, H\}$:

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h\, z_t^{(s)} + \sum_{j=1}^{p_{\text{lp}}} \rho_{h,j}\, y_{t-j} + \sum_{j=1}^{p_{\text{lp}}} \kappa_{h,j}\, \text{GPR}_{t-j}^{(s)} + \sum_{j=1}^{p_{\text{lp}}} \boldsymbol{\theta}'_{h,j}\, \text{controls}_{t-j} + \varepsilon_{t+h}. \quad (1)$$

Only the contemporaneous shock $z_t^{(s)}$ enters the RHS; GPR levels and controls enter at lags $1, \dots, p_{\text{lp}}$ only.

2.2. v3 (`ir_jorda` version)

v3 routes estimation through `ir_jorda` with `level=1`, `Jflag=1`, `I=p`, `J=0`, and *pre-expands the control matrix into p separate lag columns* (`lagmatrix(controls,1:p)`) before passing it in. This yields:

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h\, z_t^{(s)} + \sum_{j=1}^{p_{\mathrm{lp}}} \rho_{h,j}\, y_{t-j} + \sum_{j=1}^{p_{\mathrm{lp}}} \boldsymbol{\theta}'_{h,j}\, \mathrm{controls}_{t-j} + \varepsilon_{t+h}. \tag{2}$$

Net change vs. (1): v3 *removes* the GPR lag block $\sum_{j=1}^{p_{\mathrm{lp}}} \kappa_{h,j}\, \mathrm{GPR}_{t-j}^{(s)}$. All other blocks — contemporaneous shock, own-lags, and lagged controls — match v1/v2 exactly. In particular, v3 contains *no* lagged shocks, in line with the lecture-note LP variant (3).

Why these specific `ir_jorda` arguments? `Jflag=1` is required so that controls are time-aligned with the shock at *shock* time t (rather than at outcome time $t+h$, which is what `Jflag=0` would do and which would leak future information for $h > 0$). Under `Jflag=1`, the `J` argument controls the lag count of *both* the shock block ($J + 1$ columns, current + J lags) *and* the control block. To get *no* lagged shocks while still keeping p control lags, we set `J=0` and pre-expand the controls into p separate lag columns externally; `ir_jorda` then picks each pre-expanded column up at lag h, which lands the k-th column at $\mathrm{controls}_{t-k}$ inside the regression.

After the lag-expansion + `J=0` construction described above, three implementation deviations remain relative to v1/v2: the GPR lag block is dropped (statistically defensible), the HAC bandwidth is a single constant per call (`ir_jorda` restriction), and the effective regression sample is 4 observations smaller per horizon (a side-effect of how `ir_jorda` trims its head rows on top of our own contiguous-block trim).

Cross-check against "v2 minus GPR(L)". A clean robustness check is to remove the $\Gamma(L)$ GPR block from v2 (script `Q_Part2_NoVIX_v2_noGPRlag.m`) and compare its IRFs to v3's. That "v2-noGPR" variant has *exactly* the same regressor set as v3 (14 regressors per equation). The remaining gaps are the three implementation deviations of Table 1: HAC bandwidth rule, HAC sandwich code path, and the 4-observation sample gap. If v3 and v2-noGPR IRFs visually overlap, all three are immaterial in practice and our `ir_jorda` routing is validated; if they diverge materially, it pinpoints which of the three is doing the work.

Table 1. v3's spec deviations from v2/v1.

Item	**Why it changes**	**Practical implication**
Lagged GPR levels dropped	z_t is the AR(1) residual of GPR_t, so by OLS first-order conditions z_t is in-sample orthogonal to GPR_{t-1}. Under the AR(1) assumption higher GPR lags carry no extra information either. The GPR(L) block in v1/v2 was therefore largely redundant for identification of β_h. Including GPR_t (lag 0) under `Jflag=1` would also be near-collinear with z_t.	Saves p_{lp} degrees of freedom. Mild loss of robustness against *higher-order* GPR autocorrelation that the AR(1) pre-stage may not have captured; if GPR is well approximated by AR(1), the loss is essentially zero.
Single bw per call	`ir_jorda` accepts only one `L_NW` for all horizons.	Removes horizon-specific bandwidth tuning; ties bw to n_{eff}. Effect: marginally wider CI at long horizons.
Effective sample 4 obs smaller	v3 first trims to the largest contiguous in-sample block (the `lagmatrix(controls, 1:p)` pre-expansion forces NaN in rows $1..p$, so *y_block* starts at $t = p + 1$). Then `ir_jorda` internally drops another $\max(I, J) + h$ head rows at horizon h, so the first usable shock time is $t = 2p + 1$ (e.g. $t = 9$ for $p = 4$). v2's per-horizon listwise mask uses shock times $t = p + 1, \ldots, T - h$, starting at $t = 5$.	At $h = 0$: v2 $n = 156$, v3 $n = 152$; same 4-obs gap at every horizon. β_h very slightly different (small-sample) and SE slightly different. The ratio $4/156 \approx 2.6\%$ so curves should look near-identical visually.

3. Differences at a glance

Table 2. High-level differences across the three Part 2 LP scripts.

Aspect	**v1** (`Q_Part2_NoVIX.m`)	**v2** (`v2.m`)	**v3** (`v3.m`)
LP engine	In-house manual loop, horizon-by-horizon OLS	In-house, vectorised; HAC via MATLAB `hac()`	Single call to `ir_jorda` (`./code/ir_jorda.m`)
LHS	$y_{t+h} - y_{t-1}$	$y_{t+h} - y_{t-1}$	$y_{t+h} - y_{t-1}$ (via `level=1`)
RHS shock block	z_t only (no lagged z)	z_t only (no lagged z)	z_t only (no lagged z; achieved with `J=0`)
Lagged GPR levels	lags $1..p$	lags $1..p$	**dropped** (largely redundant under AR(1) shock construction; would also collide with shock at lag 0 under `Jflag=1`)
Controls (Unemp, FFR)	lags $1..p$	lags $1..p$	lags $1..p$ (no contemporaneous; achieved by pre-expanding controls into p lag columns and calling `ir_jorda` with `J=0`)
HAC bandwidth rule	$\min(h+1,\, n-1)$ (Jordà 2005 default)	$\mathrm{round}(0.18\, n_h)$ (LLSW 2018 fixed-b, per horizon)	$\mathrm{round}(0.18\, n_{\mathrm{eff}})$ (LLSW 2018 fixed-b, single bw per call)
HAC sandwich	Manual double-loop accumulation	MATLAB `hac()` (Econometrics Toolbox)	`ir_jorda` internal (uses `lrv_nw.m`)
Effective sample at $h=0$	156 (full per-horizon listwise)	156 (full per-horizon listwise)	152 (contiguous-block + `ir_jorda` head trim)
Φ^{-1} for CI	Manual `normal_icdf` (`erfcinv`)	`norminv` (Stats Toolbox)	`ir_jorda` returns *cb* directly
Lag block construction	Row-by-row scalar fill loop	Vectorised via `lagmatrix`	Internal to `ir_jorda` (`lagmatrix` based)
Data file	`.csv` only	`.mat` preferred, `.csv` fallback	`.mat` preferred, `.csv` fallback
External dependencies	None (base MATLAB)	`lagmatrix`, `norminv`, `hac` (toolboxes)	`ir_jorda`, `lrv_nw` (`./code/`), `lagmatrix`, `regress` (toolboxes)

4. Detail: HAC bandwidth comparison

All three rules are widely used Bartlett-kernel truncation lags; their implications for size and power differ.

Table 3. HAC bandwidth at typical Part 2 sample sizes ($n_{h=0} \approx 156$, $n_{h=12} \approx 144$).

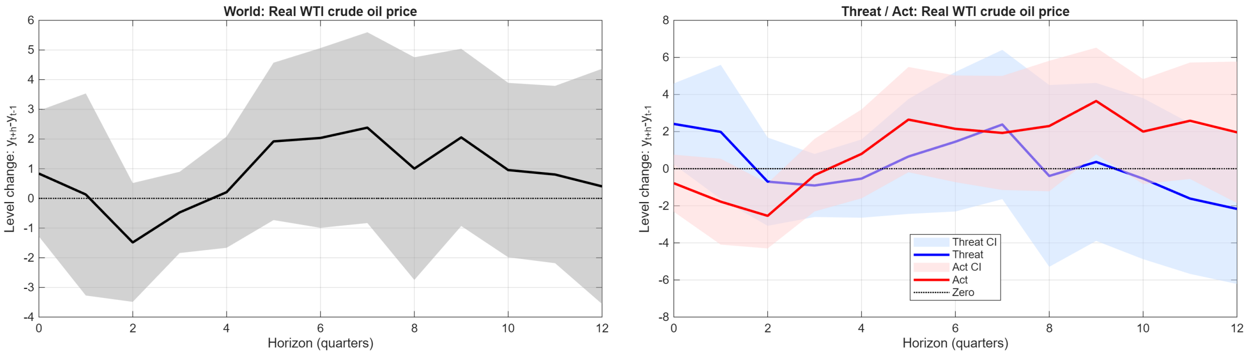
Variant	Rule	**bw** at $h = 0$	**bw** at $h = 12$	Inference target
v1	$\min(h+1, n-1)$	1	13	Standard (Newey–West) asymptotics
v2	round($0.18\,n_h$)	28	26	HAR-robust (LLSW fixed-b)
v3	round($0.18\,n_{\text{eff}}$)	28	28	HAR-robust, but single value across h

Why this matters. A wider bandwidth *lowers* the likelihood of size distortion when residuals are persistently autocorrelated (which is typical in macro LPs), at the cost of some asymptotic power. Concretely, v1 will tend to report *tighter* confidence bands than v2/v3, especially at low h; v2 and v3 intentionally widen the bands for HAR robustness. v3's bandwidth is slightly larger than v2's at long horizons because it does not shrink bw with the per-horizon sample.

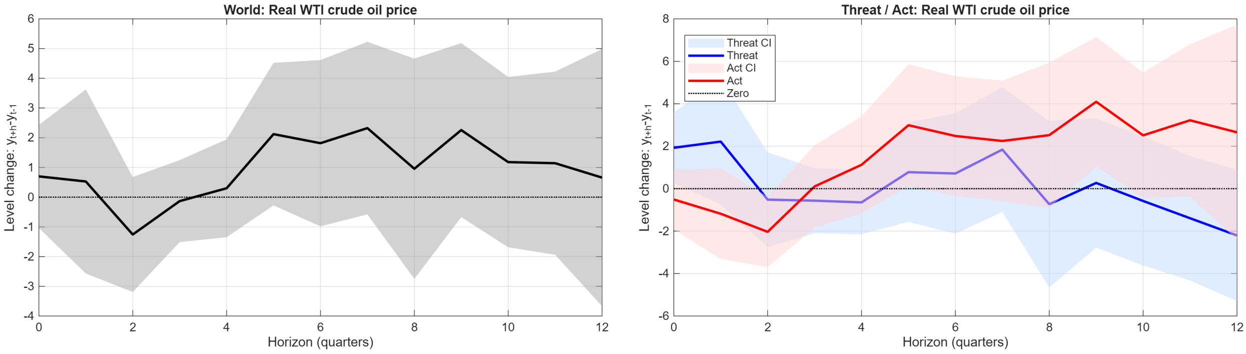
5. Figure comparison: v2 vs v3 IRFs

For each outcome we stack the v2 row strip (top) and the v3 row strip (bottom) so the two engines can be compared side-by-side at a glance. Each strip shows the *World* response on the left and the *Threat (blue) / Act (red)* responses on the right, both splitwise. **1σ shock; 90 % CI.**

5.1. Oil prices

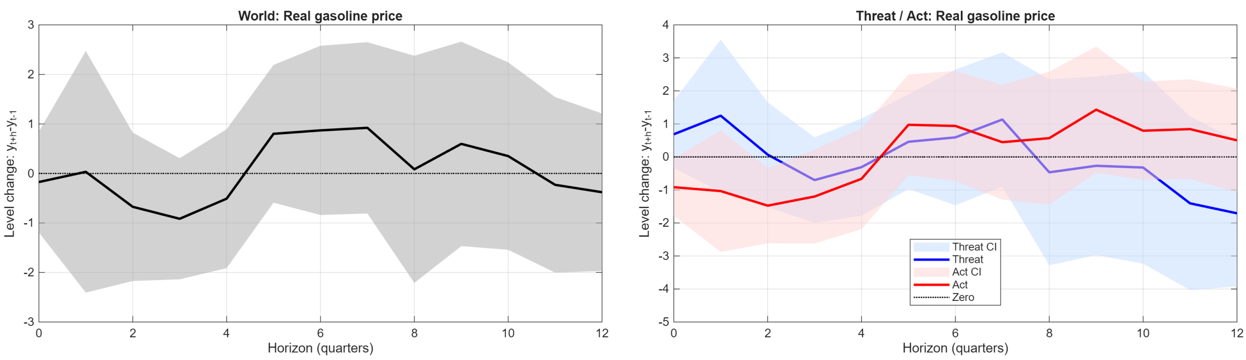


(a) v2 | manual + `hac()` | lagged GPR levels, no lagged z.

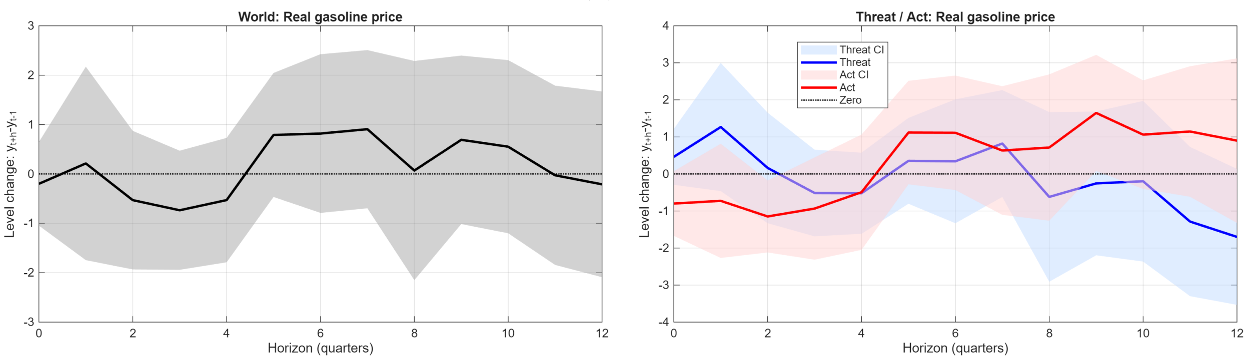


(b) v3 | `ir_jorda` | GPR-level lags dropped; otherwise identical to v2 (z_t only, controls at lags $1..p$, no contemporaneous controls).

Figure 1. Real WTI crude oil price.



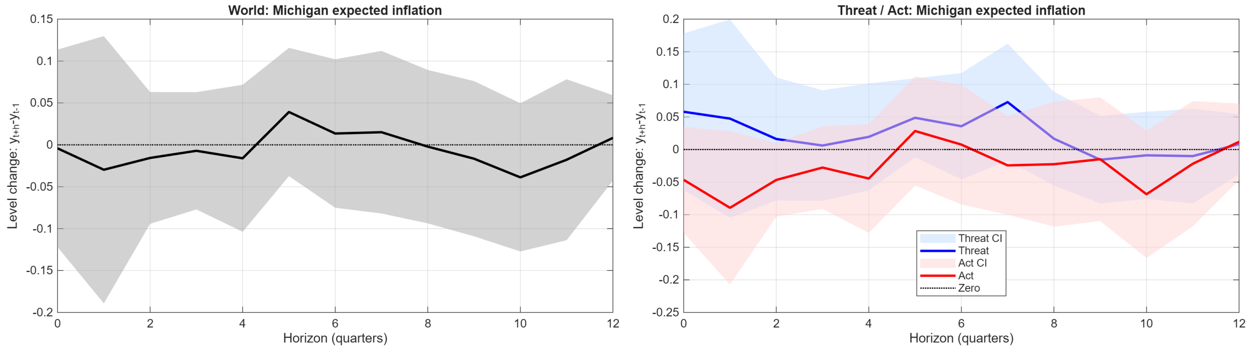
(a) v2.



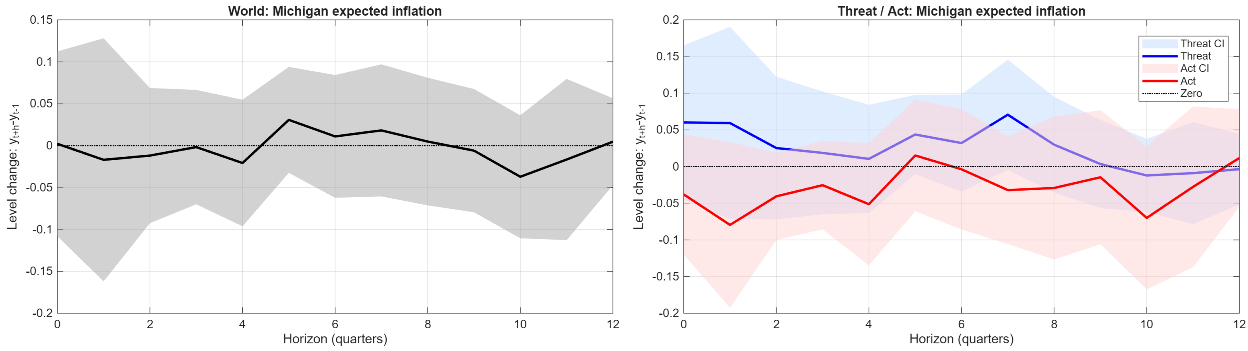
(b) v3.

Figure 2. Real gasoline price.

5.2. Inflation expectations

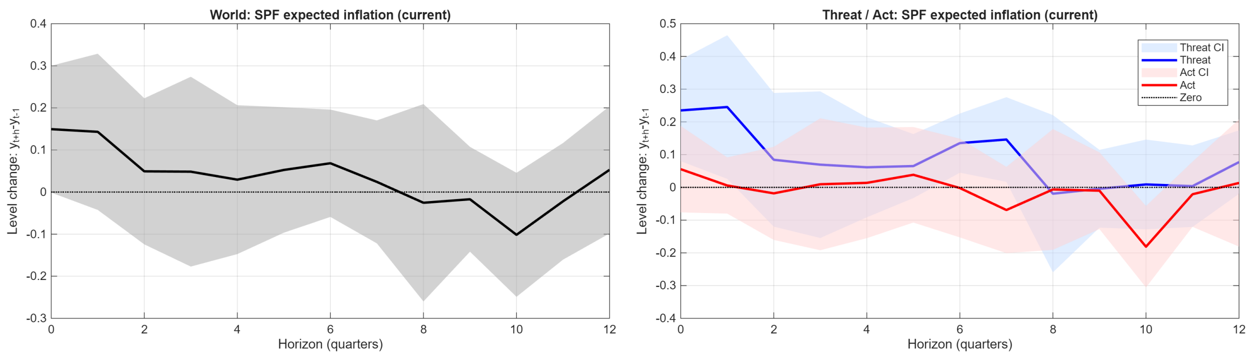


(a) v2.

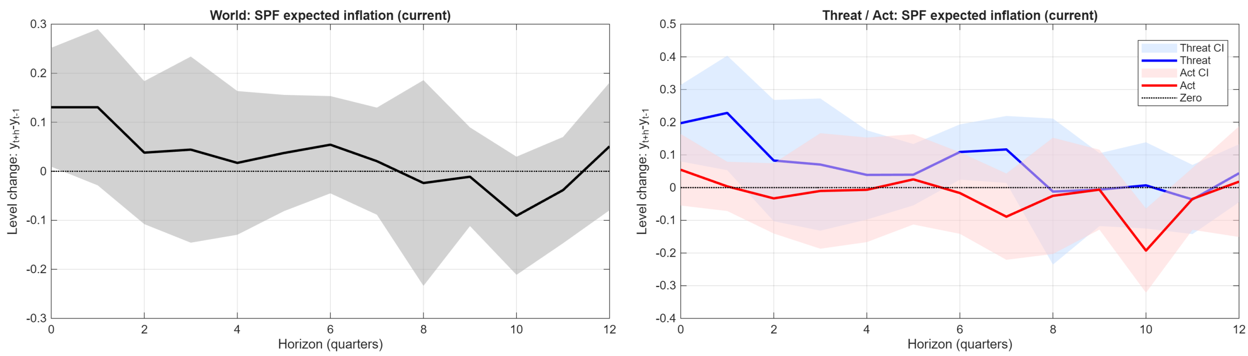


(b) v3.

Figure 3. Michigan expected inflation.

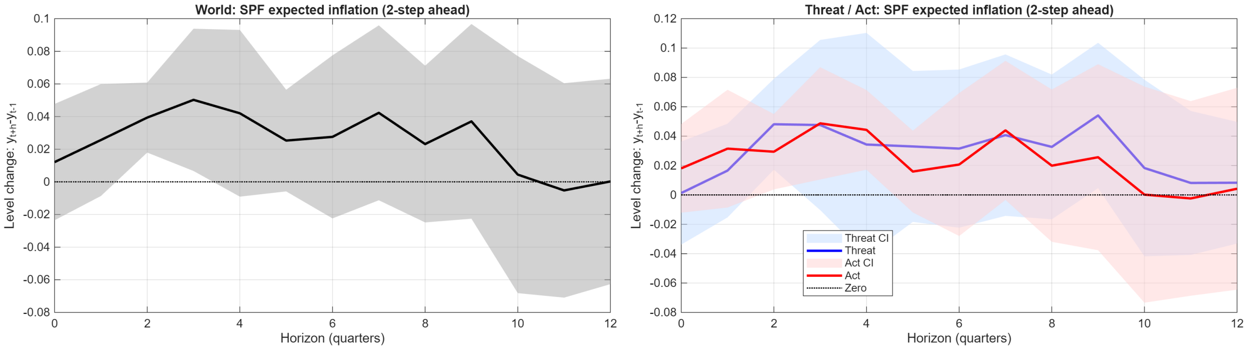


(a) v2.

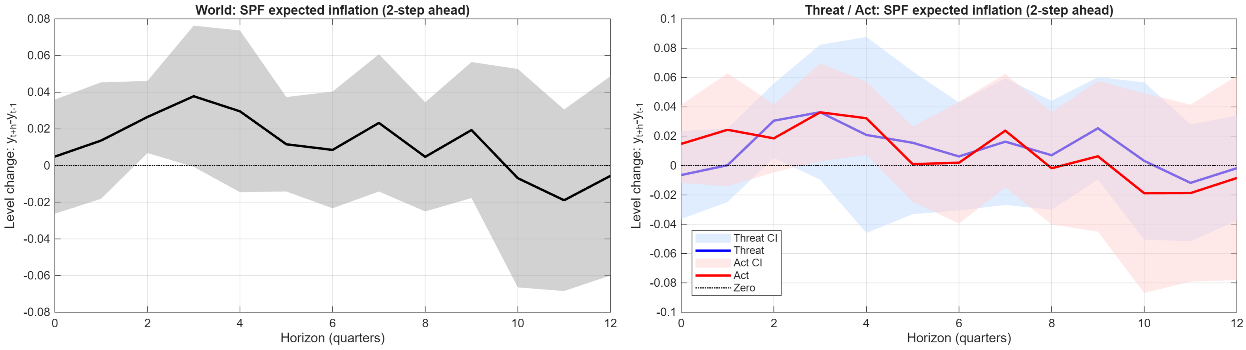


(b) v3.

Figure 4. SPF expected inflation (current).



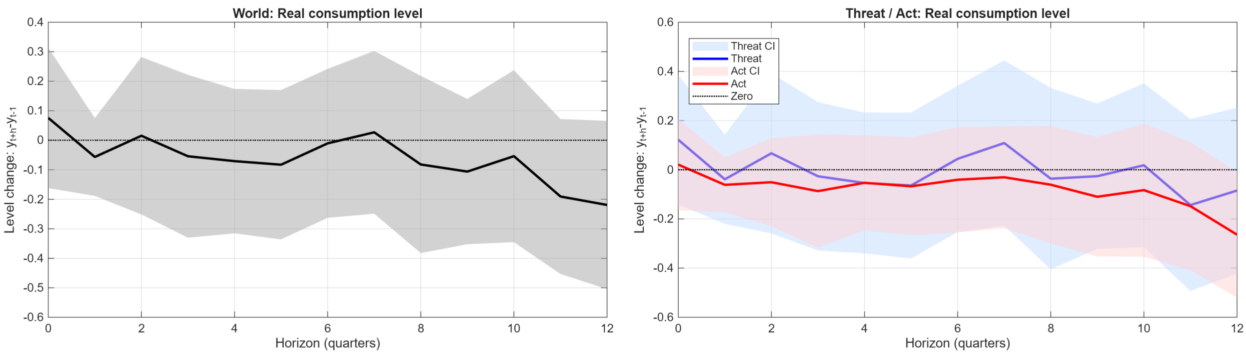
(a) v2.



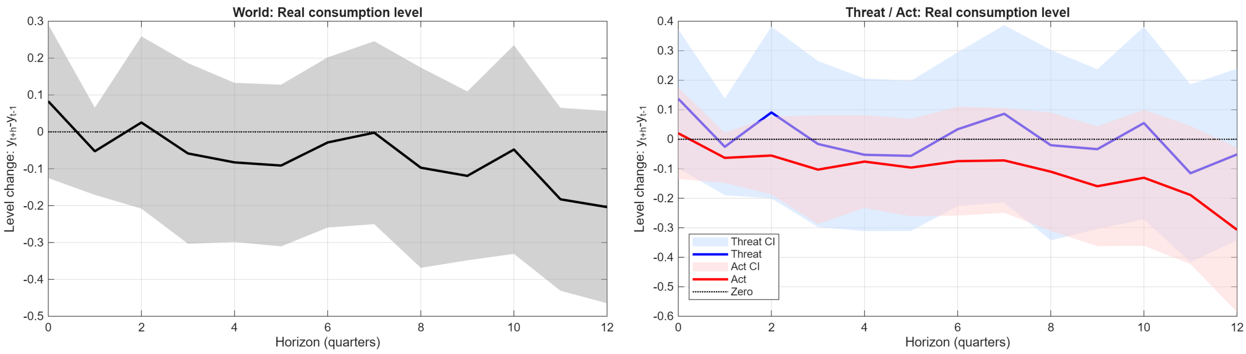
(b) v3.

Figure 5. SPF expected inflation (2-step ahead).

5.3. Real activity

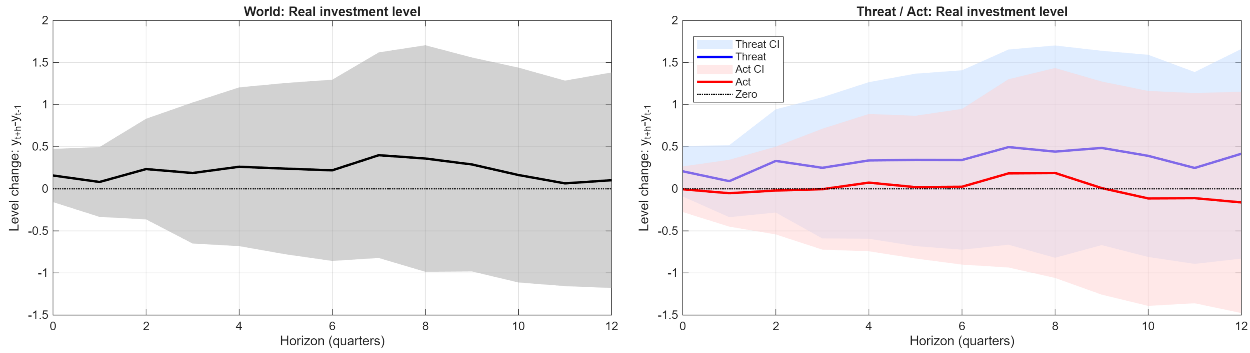


(a) v2.

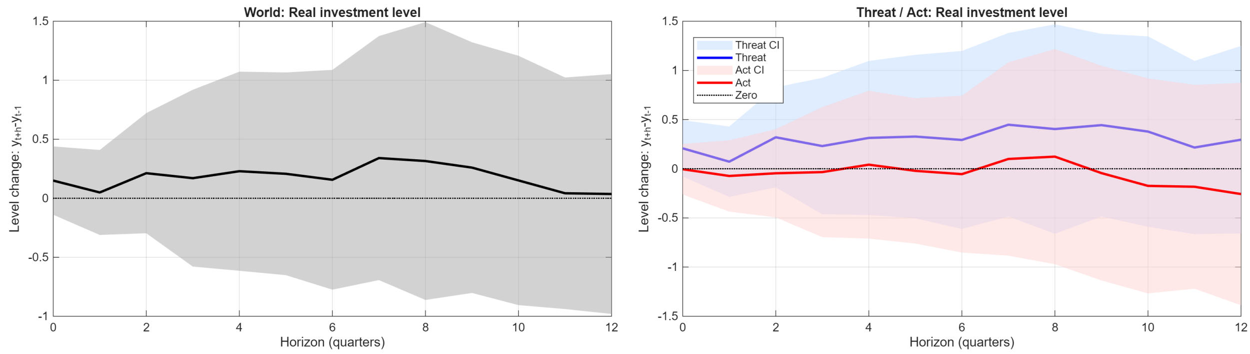


(b) v3.

Figure 6. Real consumption level.

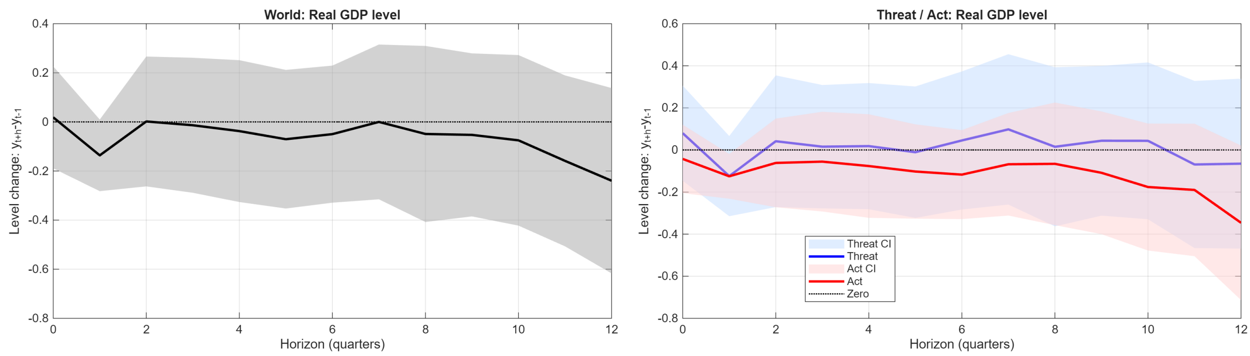


(a) v2.

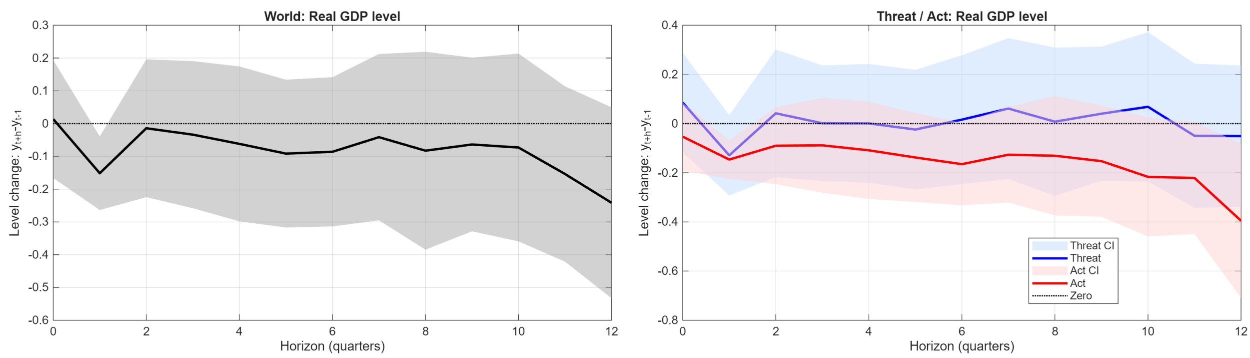


(b) v3.

Figure 7. Real investment level.



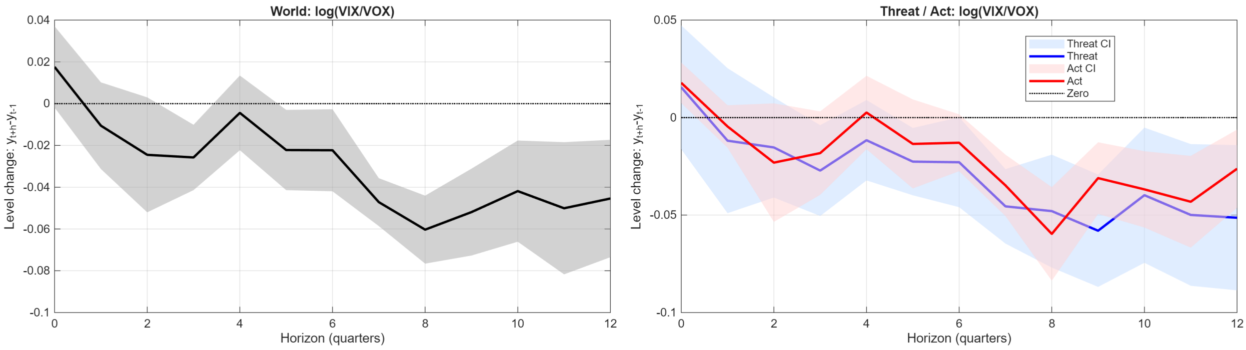
(a) v2.



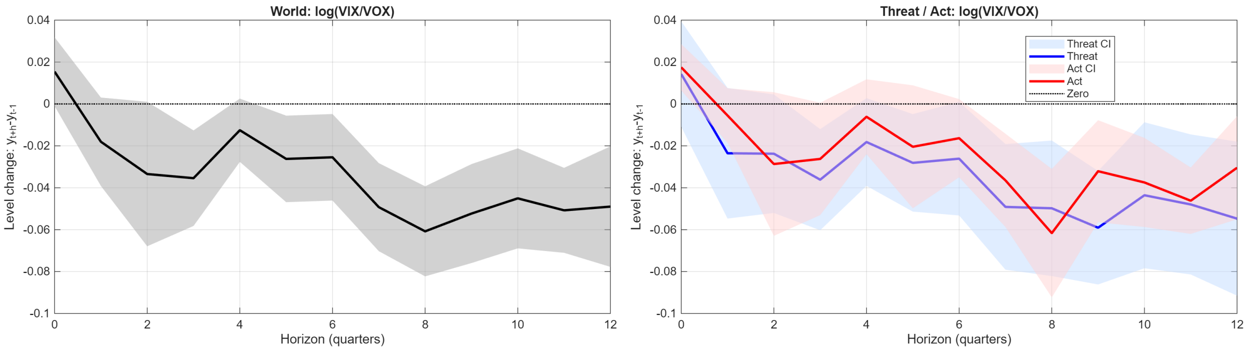
(b) v3.

Figure 8. Real GDP level.

5.4. Volatility



(a) v2.



(b) v3.

Figure 9. log VIX/VOX.